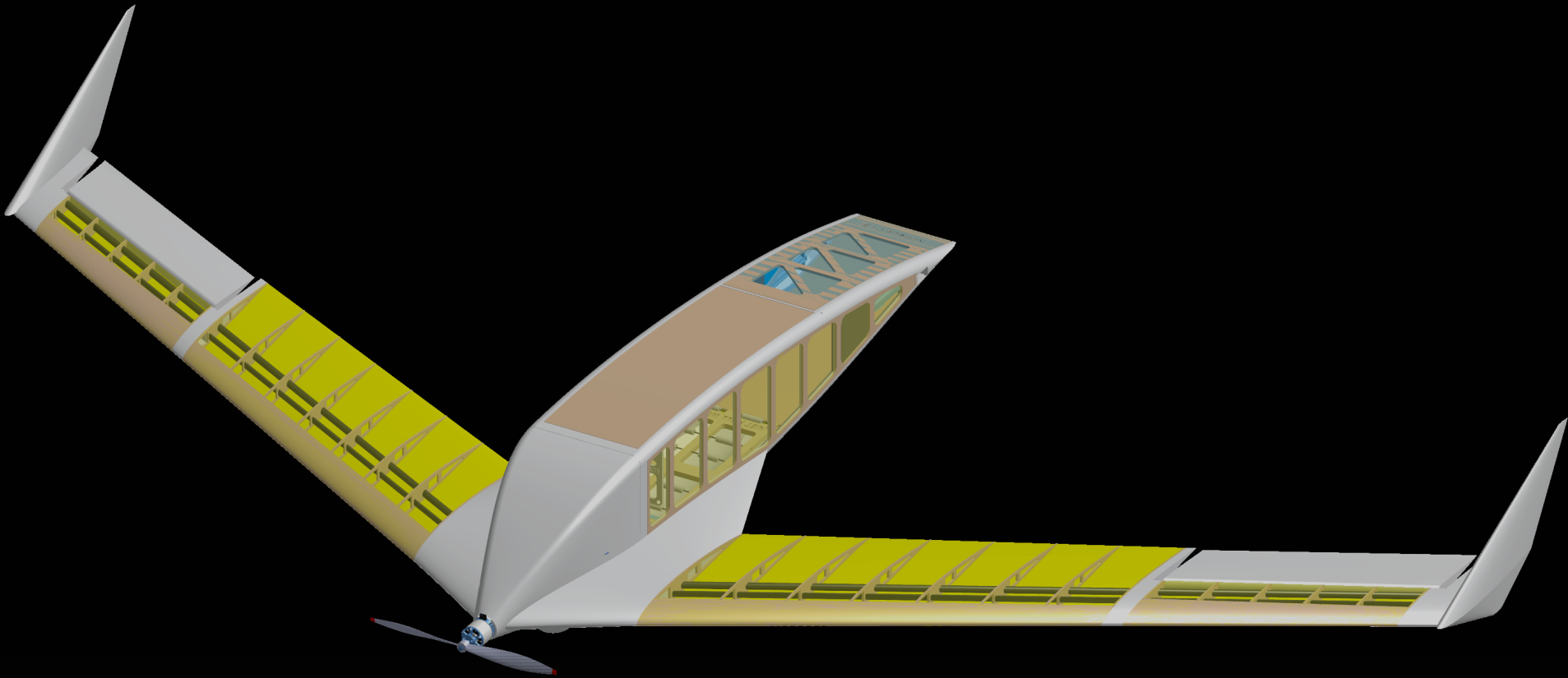


NIMBUS

PACKAGE DELIVERY UAV — SWEPT FLYING WING

MAE 155B — Aircraft Design
UC San Diego · Spring 2026
6-Person Design Team



CMA-ES optimized · XFOIL surrogate aerodynamics · AVL dynamic stability · Flight tested Spring 2026

2.60 kg

MTOW

20 m/s

CRUISE SPEED

14.34

CRUISE L/D

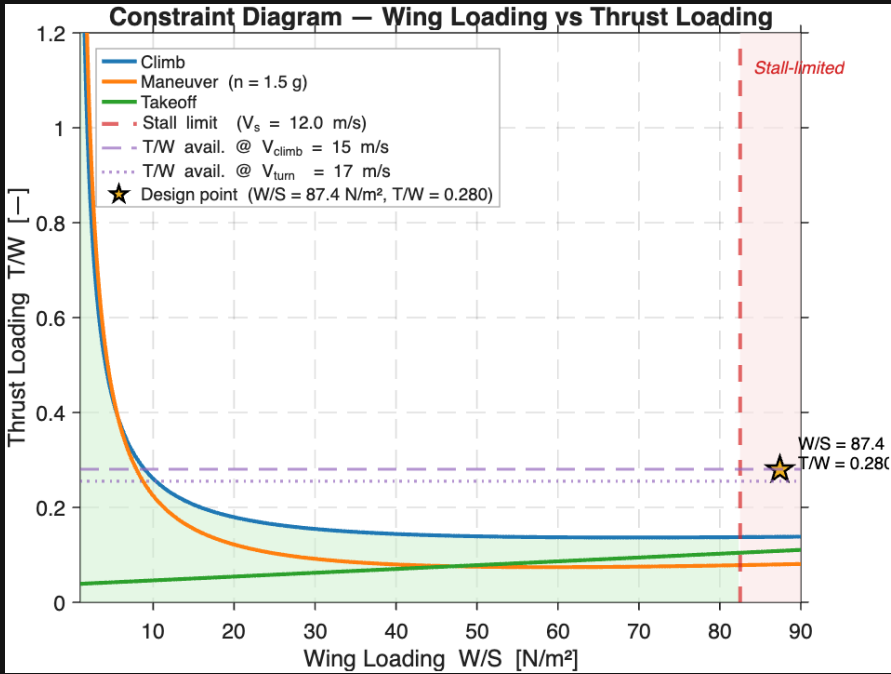
800 g

PAYLOAD

\$2.70/hr

PROFIT RATE

CTOL CONSTRAINT DIAGRAM



KEY DESIGN PARAMETERS

GEOMETRY

Wingspan	1.5715 m	61.9 in
Wing area	0.3087 m ²	—
Aspect ratio	8.0	—
Quarter-chord sweep	28.3°	—
Root / Tip airfoil	E222 / E230	—
Geometric washout	−4.04°	Panknin

WEIGHTS

MTOW (loaded)	2.60 kg	—
Payload	800 g	—

PERFORMANCE

Cruise speed	20 m/s	44.7 mph
Stall speed (loaded)	13.3 m/s	—
Wing loading W/S	87.4 N/m ²	—
Cruise L/D	14.34	—
Static thrust	10.2 N	bench-corrected

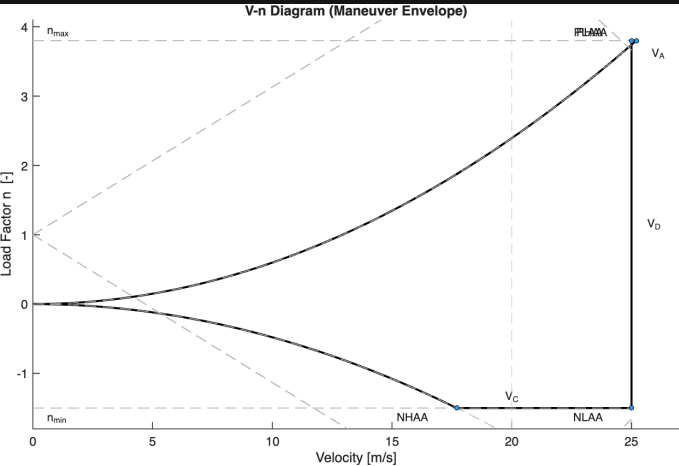
PROPULSION

Motor	SunnySky X2216 V3	1100 KV
Propeller	APC 10×4.7SF	—
Battery	3S LiPo, 11.1 V	—

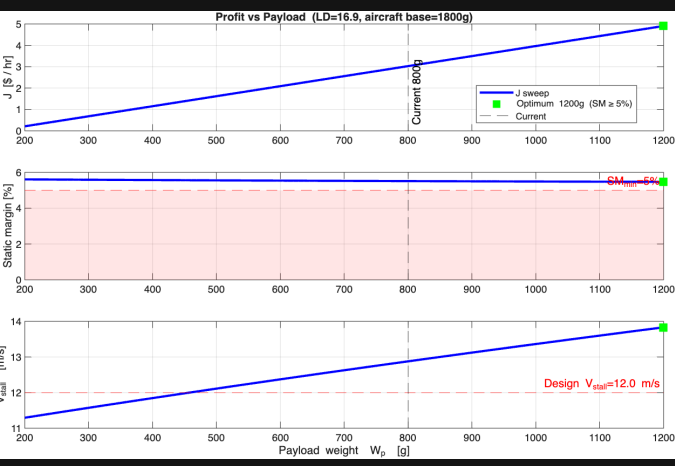
STABILITY

Static margin (AVL)	4.88%	MAC
Short-period	6.68 / 0.62	rad/s
Dutch roll	4.24 / 0.08	rad/s

V-n MANEUVER ENVELOPE



PROFIT vs PAYLOAD WEIGHT



AERODYNAMIC POLARS

